

Dipayan CHAKRABORTY

Centrale Méditerranée
Technopôle Château-Gombert
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Research domains: *identification problems in graphs, dominating sets, graph modification problems, graph algorithms, computational & parameterized complexity*

last updated : 29 mai 2026

Professional experience

- Sep. 2025 - till date **Assistant Professor (ATER in the French higher education system)**, Centrale Méditerranée, Marseille – Laboratoire d'Informatique et Systèmes (LIS), Aix-Marseille University, France
- Feb. 2025 - Aug. 2025 **Post-doctorate** (supervisor : **Faisal N. Abu-Khzam**), Department of Computer Science and Mathematics, Lebanese American University, Lebanon
- Jan. 2022 - Janv. 2025 **PhD scholar**, Laboratoire d'Informatique, de Modélisation et d'Optimisation des Systèmes (LIMOS), Université Clermont Auvergne, France
- Director **Annegret WAGLER**, Professor, Université Clermont Auvergne, France
- Co-supervisor **Florent FOUCAUD**, Associate Professor, Université Clermont Auvergne, France
- Co-supervisor **Michael HENNING**, Professor, University of Johannesburg, South Africa
- Jan. 2023 - Jan. 2025 **Research associate** (academic host : **Michael Henning**), Department of Mathematics and Applied Mathematics, University of Johannesburg, South Africa
- Oct. 2022 - Nov. 2024 **Part-time lecturer**, Institut universitaire de technologie (IUT), Université Clermont Auvergne, France
- Jan. 2021 - Dec. 2021 **Contractual researcher**, as part of the project funded by SERB-SRG and titled "Complexity dichotomies of graph modification problems", Department of Computer Science and Engineering, Indian Institute of Technology Dharwad, Karnataka, India
- Aug. 2016 - Dec. 2020 **Assistant Professor**, Department of Basic Science & Humanities, Gargi Memorial Institute of Technology, Maulana Abul Kalam Azad University of Technology, Baruiapur, Kolkata, India
- Jul. 2015 - Jul. 2016 **Visiting lecturer**, Department of Mathematics, Government College of Engineering and Ceramic Technology, Maulana Abul Kalam Azad University of Technology, Kolkata, India

Education and Trainings

- Jan. 2022 - Jan. 2025 **PhD in Computer Science** (defended 9th December 2024)
- Thesis title **Structural and algorithmic aspects of identification problems in graphs**
- Laboratoire d'Informatique, de Modélisation et d'Optimisation des Systèmes (LIMOS), Université Clermont Auvergne, France
- PhD Jury
- Director **Annegret Wagler**, Professor, Université Clermont Auvergne, France
- Co-supervisor **Florent Foucaud**, Associate professor, Université Clermont Auvergne, France
- Co-supervisor **Michael Henning**, Professor, University of Johannesburg, South Africa
- President & Examiner **Paul Dorbec**, Professor, Université de Caen-Normandie, France
- Examiner **Ralf Klasing**, Director of Research, CNRS, Université de Bordeaux, France

Reviewer **Tero Laihonon**, *Professor, University of Turku, Finlande*
Reviewer **Arnaud Pêcher**, *Professor, Université de Bordeaux, France*

Sep. 2009 - Mar. 2012 **Master in mathematics** (ALGANT Erasmus Mundus)
Université de Bordeaux, France (1st year) – Università degli Studi di Padova, Italy (2nd year)
Master thesis defended March 2012 **Mordell-Weil theorems & the Birch and Swinnerton-Dyer conjecture**, *Università degli Studi di Padova, Italy*
Supervisor **Matteo Longo**, *Università degli Studi di Padova, Italy*
Aug. 2004 - June 2007 **Bachelor of Science** (Honours in mathematics)
St. Xavier's College, Kolkata, University of Calcutta, Kolkata, India

Publications and submissions in journals

- Accepted Mar. 2026 **Tight (double) exponential bounds for identification problems : Locating-Dominating Set and Test Cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *SIAM Journal on Discrete Mathematics* (2026)
<http://arxiv.org/abs/2402.08346>
- Published Mar. 2026 **Identifying codes in triangle-free graphs of bounded maximum degree**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tuomo Lehtilä, *Journal of Graph Theory* (2026)
<https://doi.org/10.1002/jgt.70025>
- Published Mar. 2026 **Partitioning the vertex set of a graph into a dominating set and a locating dominating set**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tero Laihonon, *The Electronic Journal of Combinatorics* 33(1) : #P1.51 (2026)
<https://doi.org/10.37236/14049>
- Published June 2026 **Identifying open codes in trees and 4-cycle-free graphs of given maximum degree**, Dipayan Chakraborty, Florent Foucaud and Michael Henning, *Discrete Applied Mathematics* 386 : 319-333 (2026)
<https://doi.org/10.1016/j.dam.2026.02.032>
- Published Feb. 2026 **Identifying codes in graphs of given maximum degree : Characterizing trees**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tuomo Lehtilä, *Discrete Mathematics* 349(2) : 114826 (2026)
<https://doi.org/10.1016/j.disc.2025.114826>
- Published Jan. 2026 **On full-separating sets and related codes in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *Discrete Applied Mathematics* 379 : 551-569 (2026)
<https://doi.org/10.1016/j.dam.2025.10.001>
- Published Nov. 2025 **On open-separating dominating sets in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *Discrete Applied Mathematics* 375 : 215-238 (2025)
<https://doi.org/10.1016/j.dam.2025.05.045>
- Published Jan. 2025 **A linear algorithm for radio k-coloring of powers of paths having small diameters**, Dipayan Chakraborty, Soumen Nandi, Sagnik Sen and Supraja DK, *Journal of Computer and System Sciences* 147 : 103577 (2025)
<https://doi.org/10.1016/j.jcss.2024.103577>
- Published Dec. 2024 **Progress towards the two-thirds conjecture on locating-total-dominating sets**, Dipayan Chakraborty, Florent Foucaud, Anni Hakanen, Michael Henning and Annegret Wagler, *Discrete Mathematics* 347(12) : 114176 (2024)
<https://doi.org/10.1016/j.disc.2024.114176>
- Published Dec. 2024 **On locating and neighbor-locating colorings of sparse graphs**, Dipayan Chakraborty, Florent Foucaud, Soumen Nandi, Sagnik Sen and Supraja DK, *Discrete Applied Mathematics* 358 : 366-381 (2024)
<https://doi.org/10.1016/j.dam.2024.07.012>

- Published July 2024 **On three domination-based identification problems in block graphs**, Dipayan Chakraborty, Florent Foucaud, Aline Parreau and Annegret Wagler, *Fundamenta Informaticae (special issue : Iro Honkala's 60th birthday)*, 191(3-4) : 197-229 (2024)
<https://doi.org/10.3233/FI-242179>
- Published Jan. 2023 **On clique numbers of colored mixed graphs**, Dipayan Chakraborty, Sandip Das, Soumen Nandi, Debdeep Roy and Sagnik Sen, *Discrete Applied Mathematics* 324 : 29-40 (2023)
<https://doi.org/10.1016/j.dam.2022.08.013>
- Submitted Mar. 2025 **On the Structural Parameterizations of Locating-Dominating Set and Test Cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *Manuscript* (2025) (submitted to *Information and Computation*)
- Submitted June 2024 **The $n/2$ -bound for locating-dominating sets in subcubic graphs**, Dipayan Chakraborty, Anni Hakanen and Tuomo Lehtilä, *Manuscript* (2024) (submitted to *Graphs and Combinatorics*)

Publications and submissions in conferences with proceedings

- Accepted Apr. 2026 **On the Complexity of Vertex-Splitting Into an Interval Graph**, Dipayan Chakraborty, Faisal N. Abu-Khzam, Lucas Isenmann and Nacim Oijid, *37th International Workshop on Combinatorial Algorithms (IWOCA 2026)*
<https://doi.org/10.48550/arXiv.2602.04628>
- Published Nov. 2025 **The interplay between domination and separation in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *XIII Latin American Algorithms, Graphs, and Optimization Symposium (LAGOS 2025) Procedia Computer Science* 273 : 78-85 (2025)
<https://doi.org/10.1016/j.procs.2025.10.283>
- Published June 2025 **Structural parameterization of locating-dominating set and test cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *14th International Conference on Algorithms and Complexity (CIAC), Lecture Notes in Computer Science* 15679 : 187-204 (2025)
https://doi.org/10.1007/978-3-031-92932-8_13
- Published Feb. 2025 **On full-separating sets in graphs**, Dipayan Chakraborty and Annegret Wagler, *11th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science* 15536 :73-84 (2025)
https://doi.org/10.1007/978-3-031-83438-7_7
- Published Dec. 2024 **Tight (double) exponential bounds for identification problems : Locating-Dominating Set and Test Cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *35th International Symposium on Algorithms and Computation (ISAAC), Leibniz International Proceedings in Informatics* 322 : 19 :1-19 :18 (2024)
<https://doi.org/10.4230/LIPIcs.ISAAC.2024.19>
- Published May 2024 **Open-separating dominating codes in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *8th International Symposium on Combinatorial Optimization (ISCO), Lecture Notes in Computer Science* 14594 : 137-151 (2024)
https://doi.org/10.1007/978-3-031-60924-4_11
- Published Jan. 2024 **Location-Domination Type Problems Under the Mycielski Construction**, Silvia M. Bianchi, Dipayan Chakraborty, Yanina Lucarini and Annegret K. Wagler, *10th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science* 14508 : 255-269 (2024)
https://doi.org/10.1007/978-3-031-52213-0_18
- Published Sep. 2023 **Contracting edges to destroy a pattern : A complexity study**, Dipayan Chakraborty and Sandeep RB, *24th International Symposium on Fundamentals of Computation Theory (FCT), Lecture Notes in Computer Science* 14292 : 118-131 (2023)
https://doi.org/10.1007/978-3-031-43587-4_9

- Published Sep. 2023 **Identifying codes in bipartite graphs of given maximum degree**, with Dipayan Chakraborty, Florent Foucaud, Tuomo Lehtilä, *XII Latin-American Algorithms, Graphs and Optimization Symposium (LAGOS), Procedia Computer Science 223 : 157-165 (2023)*
<https://doi.org/10.1016/j.procs.2023.08.225>
- Published June 2023 **A linear algorithm for radio k -coloring powers of paths having small diameter**, Dipayan Chakraborty, Soumen Nandi, Sagnik Sen and DK Supraja, *34th International Workshop on Combinatorial Algorithms (IWOCA), Lecture Notes in Computer Science 13889 : 148-159 (2023)*
https://doi.org/10.1007/978-3-031-34347-6_13
- Published Jan. 2023 **On three domination-based identification problems in block graphs**, Dipayan Chakraborty, Florent Foucaud, Aline Parreau and Annegret Wagler, *9th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science 13947 : 271-283 (2023)*
https://doi.org/10.1007/978-3-031-25211-2_21
- Published Jan. 2023 **New bounds and constructions for neighbor-locating colorings of graphs**, Dipayan Chakraborty, Florent Foucaud, Soumen Nandi, Sagnik Sen and Supraja DK, *9th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science 13947 : 121-133 (2023)*
https://doi.org/10.1007/978-3-031-25211-2_9

Conferences / Summer Schools

- June 2025 **14th International Conference on Algorithms and Complexity**, (CIAC 2025)
 Organised by : Libera Università Internazionale degli Studi Sociali Guido Carli, Italy
- November 2024 **Journées Graphes et Algorithmes**, (JGA 2024)
 Organised by : Université de Bourgogne, France
- May 2024 **École de Printemps en Informatique Théorique**, (EPIT 2024)
 Organised by : École Normale Supérieure de Lyon, France
- February 2024 **10th International Conference on Algorithms and Discrete Applied Mathematics**, (CALDAM 2024)
 Organised by : IIT Bhilai, Chhattisgarh, Inde
- September 2023 **XII Latin-American Algorithms, Graphs and Optimization Symposium**, (LAGOS 2023)
 Organised by : Instituto de Matemáticas, UNAM, Juriquilla, Mexique
- June 2023 **CIMPA Research School on Graph Structure and Complex Network Analysis**, (CIMPA GSCN 2023)
 Organised by : Nesin Mathematics Village, Sirinçe, Izmir, Turquie
- February 2023 **9th International Conference on Algorithms and Discrete Applied Mathematics**, (CALDAM 2023)
 Organised by : DAIICT, Gandhinagar, Inde
- September 2022 **Maribor Graph Theory Conference**, (MGTC 2022)
 Organised by : Faculty of Natural Sciences and Mathematics, University of Maribor, Slovenie
- July 2022 **11th International Colloquium on Graph Theory and Combinatorics**, (ICGT 2022)
 Organised by : LIRMM, Montpellier, France
- june 2022 **School on Graph Theory**, *Murol, France*, (SGT 2022)
 Organised by : LIMOS, Université Clermont Auvergne, France
- May 2021 **Annual Meeting of Homomorphisms of Signed Graphs**, (HOSIGRA 2021)
 Organised by : Université de Bordeaux, Université de Montpellier et Université de Paris Diderot, France

Reviews for journals and conferences

- Conference **International Conference on Algorithms and Discrete Applied Mathematics**, *CALDAM*

International Workshop on Combinatorial Algorithms, IWOCA
Latin-American Algorithms, Graphs and Optimization Symposium, LAGOS
International Conference on Current Trends in Theory and Practice of Computer Science, SOFSEM
International Workshop on Graph-Theoretic Concepts in Computer Science, WG
International Computing and Combinatorics Conference, COCOON

Journal **Fundamenta Informaticae**
The Australasian Journal of Combinatorics

Research supervision

May 2026 - July 2025 **Intern of 3rd year bachelor's degree, IIIT Kottayam, India, Subject of research : Extremal graphs for code comparisons in identification problems in graphs**
 May 2025 - Dec. 2025 **Intern of 3rd year bachelor's degree, PSG College of Technology, Coimbatore, India, Subject of research : Identification problems in graphs**
 Co-supervised with Prof. Florent Foucaud, Université Clermont Auvergne, France
 May 2024 - July 2024 **Intern of 3rd year bachelor's degree, Chennai Mathematical Institute, Chennai, India, Subject of research : Local identification problems in graphs**
 Co-supervised with Prof. Florent Foucaud, Université Clermont Auvergne, France

Organisation of scientific events

Organising committee (Mar. 2024) **41st International Symposium on Theoretical Aspects of Computer Science, Clermont-Ferrand, France, (STACS 2024)**
 Organised by : Université Clermont-Auvergne, France

Teaching activities (in France)

Centrale Médiétrannée, Château-Gombert, Marseille, France

Sep. 2025 - till date **Computer Science Engineering : Bachelor year 1 (2025-2026 || 176 hours)**

- **JAVA** (Tutorials + Practicals || 48 hours)
 Basics of JAVA • Creating a class • Heritage & Polymorphisms • Interface • Exceptions • Reading and manipulating files
- **Algorithms** (Tutorials + Practicals || 64 hours)
 Complexities of algorithms • Algorithms on trees • Programming of algorithms on trees • Algorithms on general graphs • Dynamic programming • Algorithms on texts • Texts and lexicographic trees
- **Database** (Tutorials + Practicals || 40 hours)
 Functional dependencies and normalizations • Entity-relationship diagrams • Relational algebra • Execution of requests (SQL)

 Nov. 2025 - Jan. 2026 **Computer Science Engineering : Bachelor year 2 (2025-2026 || 18 hours)**

- **Scientific Python** (Practicals || 18 hours)
 Numpy • Matplotlib • Pandas • Seaborn • Scikit

Aix-Marseille Université, Saint-Jerôme, Marseille, France

Jan. 2025 - till date **Bachelor year 1 : Mathematics, Physics, Chemistry & Computer Science (2025-2026 || 14 hours)**

- **Programming & algorithms** (Course + Tutorials + Practicals || 30 hours)

Proving an algorithm • Iterative and recursive algorithms & pseudo-codes • Testing of codes, developing a project, and A particular project on percentages • Computational complexity of an algorithm • Complexity of a problem • Exponential Problems • Average-case complexity • Different types of sortings

Programming & algorithms (Course + Tutorials + Practicals || 30 hours)

Proving an algorithm • Iterative and recursive algorithms & pseudo-codes • Testing of codes, developing a project, and A particular project on percentages • Computational complexity of an algorithm • Complexity of a problem • Exponential Problems • Average-case complexity • Different types of sortings

Oct. 2025 - Dec. 2025 **Bachelor year 2 : Mathematics, Physics, Chemistry & Computer Science** (2025-2026 || 10 hours)

- **Graph theory and algorithms** (Practicals || 10 hours)
Class of binary trees • Huffman coding • Graphs • Computing running-times • Maximum Flow Problem (taxi of the Marne)

IUT Clermont Auvergne, Université Clermont Auvergne, Aubière, France

2022 - 2024 **Bachelor year 2 - Computer Science - Bachelor Universitaire de Technologie (BUT)** (2022-2024 || 42 hours)

- **Methods of optimisation** (Tutorials + Practicals || 42 hours)
Introduction to linear programming • Integer linear programming • Utilising linear problem solver in Sagemath • Vertex cover algorithms • Heuristic algorithms for the Travelling Salesman

Teaching activities (in India)

Gargi Memorial Institute of Technology (GMIT), MAKAUT, Kolkata, India

2016 - 2020 **Bachelor of Technology** (2016-2020 || 861 hours)

Department of Civil Engineering || Department of Computer Science Engineering || Department of Electronics & Communications Engineering || Department of Electrical Engineering || Department of Mechanical Engineering

- **Graph theory & algorithms** (Course + Tutorials || 84 hours)
- **Linear algebra** (Course + Tutorials || 120 hours)
- **Abstract algebra** (Course + Tutorials || 75 hours)
- **Probability** (Course + Tutorials || 132 hours)
- **Real analysis** (Course + Tutorials || 66 hours)
- **Complex analysis** (Course + Tutorials || 127 hours)
- **Numerical methods** (Course + Tutorials + Practicals || 143 hours)
- **Differential calculus** (Course + Tutorials || 134 hours)
- **Integral transforms** (Course + Tutorials || 64 hours)

Government College of Engineering & Ceramic Technology, MAKAUT, Kolkata, India

2015 - 2016 **Bachelor of Technology** (2015-2016 || 77 hours)

Department de Ceralic Technologie || Department de Computer Science & Information Technologies Engineering

- **Probability** (Course + Tutorials || 23 hours)
- **Integral transforms** (Course + Tutorials || 17 hours)
- **Fourier analysis** (Course + Tutorials || 17 hours)
- **Differential equations** (Course + Tutorials || 20 hours)

Academic References

- **Michael Henning**, *Department of Mathematics and Applied Mathematics, University of Johannesburg, South Africa*
mahenning@uj.ac.za
- **Sagnik Sen**, *Department of Mathematics, Indian Institute of Technology Dharwad, Karnataka, India*
sagnik@iitdh.ac.in
- **Florent Foucaud**, *LIMOS, Université Clermont Auvergne, France*
florent.foucaud@uca.fr
- **Annegret Wagler**, *LIMOS, Université Clermont Auvergne, France*
annegret.wagler@limos.fr